

Data Processing: Stony Creek, Tivoli North, and Field Station

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```
#Downloading libraries
library(readr)
library(tidyverse)
library(dplyr)
library(lubridate)
library(janitor)
```

Data and Cleaning

To obtain raw water quality and weather data, use the Centralized Data Management Office (CDMO) website: <https://cdmo.baruch.sc.edu/>

1. Navigate to the CDMO website: <https://cdmo.baruch.sc.edu/>.
2. Click on the **Get Data** tab at the top of the page.
3. Select **Advanced Query System** and click **Launch**.
4. Locate the **Zip Downloads** option and click **Choose ZIP Files**.
5. From the list of National Estuarine Research Reserves, select **Hudson River, NY**.
6. Select the appropriate dataset:
 - For **weather data** from the field station, select **hudfsmet-p**.
 - For **Stony Creek water quality data**, select **hudscwq-p**.
 - For **Tivoli North water quality data**, select **hudtnwq-p**.

Each dataset must be requested separately, as the website generates one download link per request, and the analysis code is structured to process each dataset independently.

7. After selecting the desired site, click **Submit location and proceed to next step**.
8. Select the desired range of years and click **Get files**.

- For the DA Capstone Spring 2026 project, requests were made for all available years.
9. Complete the required information requested by the website and submit the request.
 10. Shortly after submission, an email will be sent to the provided email address containing a download link to the data files.
 11. Copy and paste this link into the corresponding section of the code for the selected site.
 12. Run the code. After execution, all files will be downloaded into a subfolder named after the site within your data folder.

Note: Download link expires after certain amount of time.

Stony Creek (2002-2021)

Data Download

Data Cleaning

```
# Removing empty columns and other irrelevant columns
t <- sc_02_21_raw |>
  remove_empty("cols")|>
  select(-StationCode,
        -isSWMP, #historical and provisionalPlus columns indicate the status of QAQC
        -Historical,
        -ProvisionalPlus
  )

#creating a vector with all the flags
flags <- c("F_Record",
          "F_Temp",
          "F_SpCond",
          "F_Sal",
          "F_DO_Pct",
          "F_Depth",
          "F_cDepth",
          "F_Level",
          "F_pH",
          "F_Turb",
          "F_Ch1Fluor",
```

```
      "F_DO_mgl")

t <- t |>
  mutate(Temp = if_else(
    if_any(all_of(flags), ~ str_detect(.x, "<-3>|<-2>|<1>")), NA_real_, Temp))
```

Cleaning and creating new date and time columns

```
#the code to write out the dataset
write_csv(stony_creek, "Spring2026/data/stony_creek.csv")

#uncomment the code to write out the new/updated version of the dataset. It might be better t
```